



3D temperature field prediction in direct energy deposition of metals using physics informed neural network

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Abstract

Predicting the temperature field during the direct energy deposition (DED) process is vital for the microstructure control and property tuning of fabricated metals. The widely used data-driven machine learning method for accurate temperature prediction, however, is impractical and computation-intensive due to its sole reliance on large datasets; also being a black-box model in nature, it lacks interpretability. We propose a physics informed neural network (PINN) model, which adopts a novel physics-data hybrid method by embedding the heat transfer law into the loss function of the neural network, to model the temperature field in both single-layer and multi-layer DED. The results show that the PINN-based models with additional extrapolation ability can accurately predict temperatures with a mean relative error of 4.83%, and achieve identical prediction accuracy with only 20% of the labeled data required for training the data-driven deep neural network. The proposed model is more explainable in terms of the physics of the DED process and is also applicable for the DED of different metals.

Keywords Direct energy deposition · Temperature field · Physics-informed neural network · Physics-data hybrid machine learning

1 Introduction

DED, as a typical additive manufacturing (AM) technology, can build up three-dimensional parts by utilizing a heat source such as the laser beam to melt feedstock materials as they are being deposited. Compared to other metal AM methods, DED offers unique advantages in controlling microstructure by simply changing powder mixture and surface modification by depositing thin layers of special metal on components [1, 2]. Thus, DED has been increasingly adopted to repair damaged or worn-out components of expensive turbine blades and aerospace engines [3]. During DED processes, the temperature distribution has a great influence on microstructures and the mechanical properties of the fabricated parts. For example, different thermal

gradients and cooling rates will change the tendency to form columnar or equiaxed grain morphologies [4], which result in different micro-hardness, tensile, and fatigue resistance [5]. Moreover, the residual stress and thermal deformation are thought to be caused by the dynamic temperature distribution in DED fabricated parts [6–9]. In addition, predicting the thermal history is an important part of the digital twin for AM [10]. Therefore, it is of great significance to predict the temperature field during DED processes for acquiring the superior quality of the fabricated parts.

So far, persistent efforts of thermal prediction based on numerical methods during DED have been made. Kong et al. [11] investigated the evolution of the temperature field during the multi-layer DED process, and a level-set method was adopted to track the free surface of the molten pool by numerical modeling. An inactive or quiet elements activation method has been employed to simulate the dynamic deposition process of fed materials [12, 13]. Michaleris [14] proposed a new hybrid inactive or quiet method, which costs less time but obtains the equivalent prediction results in heat transfer simulation during the DED process. Heigel et al. [15] proposed a thermo-mechanical model of DED with measurement-based forced convection for higher accuracy. Irwin et al. [16] proposed a thermal model to predict

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the thermal history, which was further utilized to evaluate the microstructure evolution of Ti-6Al-4 V induced by the repeated thermal cycling during DED processes. In [17], the temperature collected by an IR camera was used to calibrate the input parameters, which improved the prediction accuracy of the numerical model. The numerical simulation models above have greatly improved in accuracy, stability, and efficiency in simulating the temperature distribution of DED. However, it is difficult to meet the demands of real-time prediction in real AM [18] using numerical methods. In addition, developers are required to not only understand the laws of physics but also master many mathematical skills. Besides, computing a high-fidelity model usually requires high-performance hardware and long computation time, which are computationally expensive [19].

In addition to the numerical method, data-driven methods have also been used for modeling the temperature field during DED due to their powerful ability of nonlinear regression and classification. Stathatos and Vosniakos [20] developed an artificial neural network (ANN) to predict the evolution of temperatures for long tracks in DED. In another literature [21], an IR camera was applied to capture the peak temperature of the molten pool under different processing parameters in real time. Based on the data collected, extreme gradient boosting (XGBoost) and long short-term memory (LSTM) were used to establish the relationship between the processing parameters and the molten pool temperatures. Both of the two algorithms have been proved to be capable of achieving high accuracy in predicting the molten pool temperature during the DED process. However, the thermal models above can only predict the peak temperature of the molten pool but cannot predict the temperature distribution across the whole part. Mozaffar et al. [22] proposed a thermal history prediction model based on a recurrent neural network (RNN). The model was trained with high volumes of data generated by the finite element method (FEM) of DED, so it can predict temperature fields not only on the surface but also within the interior of fabricated parts under different parameters and arbitrary geometries in DED. Ren et al. [23] trained a combined RNN-DNN model with sufficient data generated by numerical simulation to identify the correlation between laser scanning strategies and thermal history distributions during the DED process. A novel data structure was developed to describe the deposition state. The result of the developed model was in good agreement with the numerical simulation result.

Even though some temperature field prediction models in DED processes have been established based on data-driven machine learning, there are still some research gaps and challenges. Firstly, the data-driven model, which is black-box in nature, is established through pure data fitting with no or limited physical knowledge and cannot be interpreted

from the perspective of physical principles. Secondly, the prediction accuracy of the data-driven model largely depends on the amount and representation quality of the labeled data. The model always performs poorly in situations where the training data does not cover it. In order to achieve high accuracy and good robustness, a large number of labeled data obtained under different conditions are required to train the model. However, it is expensive and time consuming to acquire data by experiments or numerical simulations and then to label the data for machine learning [24, 25]. Recently, PINN [26], a novel physics-data hybrid machine learning method in which the learning process conforms to some specific laws of physics defined by some partial differential equations, has been proposed to solve problems in crack identification [27], subsurface transport [28, 29], electromagnetic analysis [30], cardiovascular flows [31], and material design [32]. Here, we employ this physics-data hybrid machine learning method that integrates the interpretability of numerical simulation and the predictive capabilities of the machine learning algorithm to model the temperature field in DED processes. Since the temperature field is the result of heat transfer that can be well described by the heat conduction partial differential equation (PDE), the main idea of this work is to embed the heat conduction PDE into the neural network so that the established temperature field prediction model is not only driven by data but also constrained by the heat conduction laws simultaneously. Our proposed approach of modeling the 3D temperature field using the physics-data hybrid method makes a novel contribution to the field of intelligent materials processing, specifically DED. First, the established model using PINN can accurately predict the 3D temperature field during the DED process. Second, the trained model is explainable with the physical laws of the DED thermal process, in contrast to data-driven machine learning models which are black-box in nature. Third, due to additional constraints enforced by the physical laws, the learning process requires significantly less labeled data than data-driven machine learning models based on DNN or LSTM.

The remainder of this paper is organized as follows. Section 2 describes the governing laws of heat transfer during DED processes and presents the hybridized modeling process by combining the neural network with the physical laws. Section 3 describes the method of generating data by FEM, which has been proven feasible [19, 23]. Besides, DED experiments are designed to validate the reliability of the data generated by FEM. Section 4 presents the temperature prediction results using PINN in several cases and the experimental validations. Section 5 discusses the physical parts in the loss function of the established PINN, then compares the performance of the physics-data hybrid method with that of the data-driven methods. Finally, Sect. 6 makes conclusions and some suggestions for future work.

2 Methodology

2.1 An overview of PINN

In finding the solution of a complex nonlinear relationship between the input $[x, t]$ and output u , we generally use a data-driven method such as training a fully connected neural network to map the input and output (see Fig. 1a). The parameters of the neural network are updated through forward propagation and backpropagation. In the process of backpropagation, gradient descent is used to optimize the parameters of the neural network so as to minimize the loss function value, which represents the discrepancy between the predicted value and the real value in supervised learning. The prediction accuracy of the data-driven model largely depends on the amount and representation quality of the labeled data. However, in real physical scenarios, there are usually some underlying physical laws between $[x, t]$ and u , and the physical laws can be described by the relevant PDEs. If the physical laws can be utilized to constrain the learning process, the model will not only be driven by data but also be guided by the physical laws.

The newly proposed PINN method incorporates the data and physical laws to train the neural network. Specifically, provided that the input (x, t) and the output u satisfy the general form of a physical PDE:

$$\frac{\partial u}{\partial t} + N[u; \lambda] = 0, x \in \Omega, t \in [0, T] \quad (1)$$

where $u(x, t)$ is the target mapping between the input and output, and it is also the latent solution of the PDE. $N(u; \lambda)$ is a nonlinear differential operator, λ represents the parameters in the PDE, Ω is the domain of the system, and $[0, T]$ is the time interval. To embed the physical law into the neural network, we use $r(x, t)$ to represent the residual of the PDE:

$$r(x, t) = \frac{\partial u}{\partial t} + N[u; \lambda] \quad (2)$$

If the predicted output $u_{NN}(x, t)$ of the neural network approaches the solution $u(x, t)$, $r(x, t)$ will approach 0, indicating that the mapping between the input and output satisfies the physical law described by the PDE. $r(x, t)$ is

calculated by computing the partial derivatives of the predicted $u_{NN}(x, t)$ with respect to the input x and t . As a result, the neural network predicting $u(x, t)$ shares the same parameters with the neural network predicting $r(x, t)$. The shared parameters of the two neural networks can be optimized by minimizing the newly designed loss function:

$$L = L_u + L_r \quad (3)$$

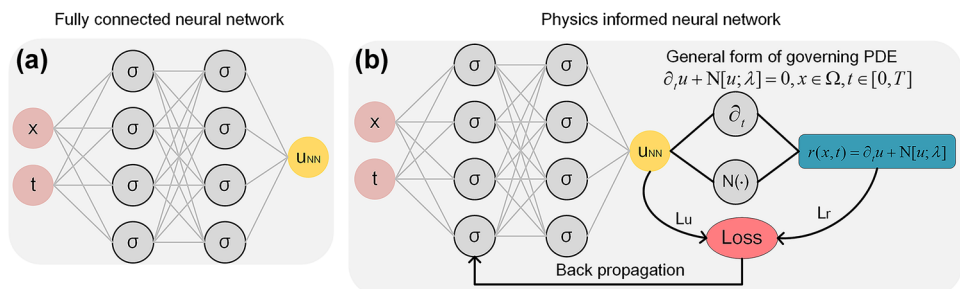
$$L_u = \frac{1}{N_u} \sum_{i=1}^{N_u} |u(x_u^i, t_u^i) - u^i|^2 \quad (4)$$

$$L_r = \frac{1}{N_r} \sum_{i=1}^{N_r} |r(x_r^i, t_r^i)|^2 \quad (5)$$

where L_u is the mean squared error loss between the predicted value and the real value and L_r is the mean squared error of the PDE residual that penalizes the discrepancy between the predicted solution and the real solution of the physical law. The schematic diagram of PINN is shown in Fig. 1b.

The PINN-based model has much clearer interpretability with physics meaning than data-driven methods. In data-driven methods, the nonlinear relationship between the input and the output of the model is fitted by a large amount of data without taking into account the underlying physical laws in the usage scenarios. However, in PINN, the PDE that represents the physical laws is embedded into the loss function to guide the solution between the input and output to approach the feasible solution of the PDE. Specifically, the aim of each backpropagation is to minimize the loss function to approach 0. If the input and the predicted value deviate from the feasible solution of the heat conduction PDE, the absolute value of the PDE residual will be magnified and the loss function will be magnified. Then the gradient descent optimizer will update the parameters of the neural network to make the predicted value approach the solution of the PDE. In a word, the learning process of the PINN model is guided and constrained by the heat conduction laws in DED, and thus the PINN model is not a fully data-driven model.

Fig. 1 The schematic diagrams of **a** the fully connected neural network and **b** the general form of PINN. σ represents the activation function in the neural network



2.2 Governing physical laws of heat transfer during the DED process

During the DED process, the transient heat transfer can be defined by the following heat conduction PDE [5]:

$$\frac{\partial(\rho c T)}{\partial t} - \left[\frac{\partial}{\partial x} \left(k \frac{\partial T}{\partial x} \right) + \frac{\partial}{\partial y} \left(k \frac{\partial T}{\partial y} \right) + \frac{\partial}{\partial z} \left(k \frac{\partial T}{\partial z} \right) \right] = 0 \quad (6)$$

where, T is the temperature ($^{\circ}\text{C}$), as a function of time t and spatial coordinates x , y , and z , ρ is the density (kg/m^3) of the deposited materials, c is the specific heat ($\text{J}/(\text{kg}\cdot^{\circ}\text{C})$) of materials, and k is the thermal conductivity ($\text{W}/(\text{m}\cdot^{\circ}\text{C})$) of materials. The values of ρ , c , and k are thought to change with the temperature. In the initial state ($t=0$ s) and at the boundary of the computational domain, the temperature field satisfies the following conditions:

$$T(x, y, z, 0) = T_0(\text{initialcondition}) \quad (7)$$

$$T(t, x, y, z) = T_{bc}, (t, x, y, z) \in \Omega(\text{boundarycondition}) \quad (8)$$

where T_0 is the initial temperature field, T_{bc} is the temperature on the boundary, and Ω represents the temporal and spatial coordinates on the boundary. Thus, the governing physical laws of heat transfer during the DED process are described by (Eqs. 6–8).

2.3 Modeling process using PINN

The forward propagation of PINN is similar to that of a fully connected neural network which is built to map the input and output [26]. Compared to data-driven machine learning, the core idea of PINN is to train neural networks with the governing PDE and given data [33]. To embed the physical laws into the neural network, in PINN, the residual of the PDE is worked as a part of the loss function, which enables the learning to be constrained by the physical laws. In real DED manufacturing, the thermal parameters ρ , c , and k of materials in Eq. 6 are not static but temperature-varying, leading to high complexity in modeling them, respectively. Instead, we use the diffusion coefficient α to reflect the heat transfer ability of the material (Eq. 9). The real value of α is difficult to be determined through experimental measurement or defined by an explicit function. Even if α can be defined by an approximate function, the function is thought to be either complex or not accurate enough to reflect the real heat diffusion state during the DED process. Once inaccurate values or functions are embedded into the neural network, the gradient explosion would occur and the established model would not converge. Therefore, similar to weights and biases in the neural network, α will be treated as a

parameter to be updated dynamically by a data-driven method for acquiring desired prediction accuracy.

$$\frac{\partial T}{\partial t} - \alpha \cdot \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} \right) = 0 \quad (9)$$

To embed the physical laws into the neural network, R_T called the residual of the PDE, represents the left part of Eq. 9:

$$R_T = \frac{\partial T}{\partial t} - \alpha \cdot \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} \right) \quad (10)$$

where T is the output of each iteration of the neural network. When T matches the real temperature that conforms to the PDE (Eq. 9), the value of R_T will approach 0 and Eq. 10 will be equivalent to Eq. 9, indicating that the predicted temperature field satisfies the PDE of the heat conduction. Therefore, the converging process of R_T is consistent with the process that the predicted temperatures approach the labeled data. Inspired by this, R_T will be embedded into the loss function as an additional constraint of the learning process. Taking the initial condition and boundary condition into account, the loss function consists of four parts: (1) data constraint L_{data} , (2) physical laws constraint L_{physics} , (3) initial condition constraint L_{init} , and (4) boundary condition constraint L_{boundary} . In this paper, the loss function is defined as follow:

$$L = L_{\text{data}} + L_{\text{physics}} + L_{\text{init}} + L_{\text{boundary}} \quad (11)$$

$$L_{\text{data}} = \frac{1}{m} \sum_{i=1}^m (\hat{T}^{(i)} - T^{(i)})^2 \quad (12)$$

$$L_{\text{physics}} = \frac{1}{m} \sum_{i=1}^m R_T^2 \quad (13)$$

$$L_{\text{init}} = \frac{1}{m_0} \sum_{i=1}^{m_0} (\hat{T}_0^{(i)} - T_0^{(i)})^2 \quad (14)$$

$$L_{\text{boundary}} = \frac{1}{m_b} \sum_{i=1}^{m_b} (\hat{T}_b^{(i)} - T_b^{(i)})^2 \quad (15)$$

where $\hat{T}^{(i)}$ is the output of the model, $T^{(i)}$ is the labeled temperature, $\hat{T}_0^{(i)}$ is the output of the model at $t=0$ s, $T_0^{(i)}$ is the labeled temperature at $t=0$ s (initial condition), $\hat{T}_b^{(i)}$ is the output of the model at borders of the computation domain, and $T_b^{(i)}$ is labeled temperature at the borders of the computation domain (boundary condition).

The computation flow of the PINN model is shown in Fig. 2. The inputs of the model include the laser power (p), scanning speed (v_s), time (t), and spatial coordinates x , y , and z . After forward propagation, the model outputs the

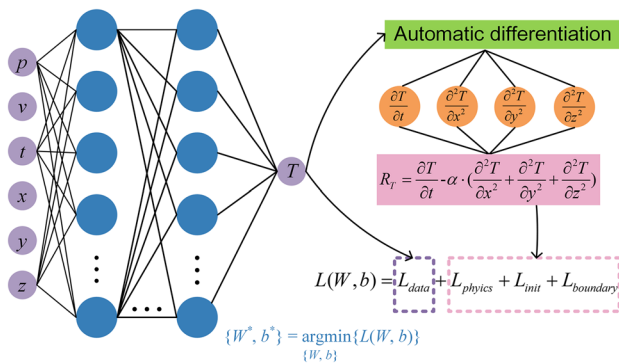


Fig. 2 Schematic diagram of the PINN model for predicting the temperature field of DED

predicted temperature. To obtain the result of the loss function in the backpropagation process, automatic differential (AD) [34] is used to calculate the partial derivatives of T with respect to t , x , y , and z , respectively. AD can calculate the partial derivatives since the function between the input and output is established. Finally, optimizers are utilized to minimize the loss function by updating the parameters (weights, biases, and α) in the neural network through iterative backpropagations.

We have conducted an elaborate investigation to determine the appropriate number of hidden layers, the number of neurons per layer, and the learning rate with 5000 training data and 2000 testing data. As shown in Table 1, the grid search method is employed to obtain the optimal combination of the hidden layer and the neurons per layer of the PINN. The prediction error is evaluated by the relative L_2 norm error, $e = \|\hat{y} - y\|_2 / \|y\|_2$, where $\hat{y}$ is the predicted value and y is the labeled value. Learning rate is an important hyperparameter which affects the convergence speed and the prediction accuracy. A set of learning rates are tested (see Table 2) to find an appropriate value. Consequently, 5 hidden layers with 50 neurons per layer and the learning rate 0.0005 are set as the applicable hyperparameter configuration. The output of the previous layer works as the input of the next layer, which can be described as

$$a_i = \tanh(W_i * a_{i-1} + b_i) \quad (16)$$

Table 1 The prediction error (10^{-2}) for different number of hidden layers and neurons per layer (the learning rate is 0.0005)

Neurons Layers	30	40	50	60
3	4.72	4.84	4.82	5.01
4	5.39	5.11	5.16	5.46
5	7.73	5.97	2.61	6.73
6	2.62	2.56	5.38	2.72

Table 2 The prediction error for different learning rates (5 hidden layers with 50 neurons per layer)

Learning rate	0.0001	0.0005	0.001	0.005	0.01
Error (10^{-2})	4.93	2.61	3.90	14.9	47.8

where a_{i-1} is the output of the $i-1$ th layer, a_i is the output of the i th layer, W_i represents the weight matrix of the i th layer, and b_i represents the bias matrix of the i th layer. The activation function is set as the hyperbolic tangent function ($\tanh$ in Eq. 16). Compared with the rectified linear unit (ReLU) activation function, the $\tanh$ function is a smooth and zero-centered function thereby boosting the backpropagation process. Besides, the convergence speed of the $\tanh$ function is faster than the sigmoid function [35]. Adam optimizer, which is easy to implement and requires small memory, combines the advantages of Adagrad optimizer and RMSProp optimizer and is able to deal with sparse gradients and non-stationary objectives [36]. The limited memory BFGS (L-BFGS) method, with a fast convergence speed, can optimize the neural network continuously until the absolute difference between the consecutive two-step loss functions is less than 10^{-8} [37]. In our case, the loss function is firstly minimized by the Adam optimizer for 30,000 steps and further optimized by the L-BFGS optimizer for the subsequent 10,000 steps.

Considering that the loss function L can be given by different combinations of Eqs. 12–15, we also examine the prediction error for the loss function in forms of L_{data} , L_{physics} , $L_{\text{data}} + L_{\text{physics}}$, $L_{\text{data}} + L_{\text{init}} + L_{\text{boundary}}$, and $L_{\text{physics}} + L_{\text{init}} + L_{\text{boundary}}$. We have debugged the optimal hidden layer number, the neuron number per layer, and the learning rate for each experiment using the grid search method and other experiment setups (e.g., the number of training data, optimizers, iteration numbers et al.) keep the same. The final results are summarized in Table 3, where the result for the loss function in the form of $L_{\text{data}} + L_{\text{physics}} + L_{\text{init}} + L_{\text{boundary}}$ (see Eq. 11) is defined by the PINN model is also presented for comparison. The results indicate that the model using the loss function only with the physical term L_{physics} fails to accurately predict the temperature field because the PDE cannot be solved directly without sufficient boundary conditions. And the obtained prediction error is reduced when the boundary condition L_{boundary} and initial condition term L_{init} are added into the loss function. When the data term L_{data} is further considered in the loss function (see Eq. 11), the performance is found to be the best, demonstrating that our proposed PINN model is based on the hybrid method incorporating the physical law and data has a superior performance.

Table 3 The prediction errors for the loss function in different forms

The loss function form	L_{data}	L_{physics}	$L_{\text{data}} + L_{\text{physics}}$	$L_{\text{physics}} + L_{\text{init}} + L_{\text{boundary}}$	$L_{\text{data}} + L_{\text{init}} + L_{\text{boundary}}$	$L_{\text{data}} + L_{\text{physics}} + L_{\text{init}} + L_{\text{boundary}}$
Error (%)	7.52	82.6	5.72	6.68	5.88	2.61
Learning rate	0.0005	0.0005	0.0005	0.0005	0.0005	0.0005
Hidden layer	5	2	5	5	4	5
Neurons	50	50	50	40	30	50

3 Data preparation and experimental setup

Infrared cameras and pyrometers are usually used to monitor the temperatures on the surface of the fabricated parts by converting radiation to electric signals [38]. However, they are not accessible to the temperatures inside the parts. Moreover, it is difficult to identify the corresponding spatial coordinate of the captured temperature. To obtain reliable and enough 3D temperature field data of the fabricated parts, the high-fidelity FEM was adopted to simulate the heat transfer during the DED process in the platform of ANSYS Workbench 19.0. After the results of FEM were validated by experiments, the data generated by simulation was used to train the PINN based models. Table 4 illustrates the parameter settings for FEM. The values of density, thermal conductivity and specific heat of Inconel 718 are imported from the engineering data source of Workbench 19.0 and the subscripts of the density, thermal conductivity and specific heat represent the corresponding temperatures (e.g., 8220₂₀ means that the density is 8220 kg/m³ under 20 °C). The chemical composition of Inconel 718 is listed in Table 5. Considering the energy of the laser heat source conforming to the Gaussian distribution, the heat source can be expressed as

$$q(x, y, t) = \frac{3\eta p}{\pi r^2} \exp\left(-3 \times \frac{(x - x_0 - v_s t)^2 + (y)^2}{r^2}\right) \quad (17)$$

where p is the laser power, r is the effective radius of the laser beam (0.8 mm in this study), η is the laser absorption coefficient (0.45 for Inconel 718 [39]), and v_s is the scanning speed. The emissivity of Inconel 718 is set as 0.5 and the convection coefficient is set as 25.0 W/m² in the FEM [40].

Figure 3(a) depicts the overall DED experiment system. The six-axis robot (KUKA, KR30HA) equipped with coaxial

nozzles is employed to implement the designed path. The laser radiation is generated by a 2000 W fiber laser (IPG, YLS-2000-CT). The metal powder is supplied by argon gas. The powder feeding rate is controlled by a powder feeder (JIU-CHUN (Suzhou) Co, Ltd). The molten pool is also shielded by argon gas. As shown in Fig. 3(b), an infrared bicolor thermometer (METIS M3) that can capture the peak temperature within the range of laser spot is used to measure the molten pool temperature. In the bicolor infrared thermometer, two detectors with similar response wavelengths can collect the heat signals synchronously. There is a fixed relationship between the ratio of the two collected signals and the measured temperature, so the measured temperatures will not be affected by the emissivity that is difficult to be determined accurately. Thus, the measured temperatures are more reliable. The schematic diagram of temperature measurements is shown in Fig. 3(c).

The reliability of the predicted temperature field using the PINN method relies on the fidelity of the data generated by the FEM. In this paper, DED experiments were performed to validate the results of FEM. The peak temperatures of the molten pool are obtained under the laser power of 240 W, or 260 W, corresponding to a scanning speed of 5 mm/s or 4 mm/s, respectively. As shown in Fig. 4, the temperatures calculated by FEM are in good agreement with those by experimental measurements, demonstrating that the temperature data from FEM are reliable and can be used as training data.

4 Results

In this paper, PINN is applied to model the temperature field in both single-layer DED and multi-layer DED to validate the effectiveness of the proposed method in different DED processes. With the data generated by FEM, 6000 samples

Table 4 Simulation parameters

Name	Settings							
Density (kg/m ³)	8220 ₂₀	8048 ₄₂₇	7875 ₉₂₇	7733 ₁₂₆₀	7579 ₁₃₄₄	7488 ₁₄₅₀	7488 ₁₅₂₇	7341 ₁₈₂₇
Thermal conductivity (W/(m•°C))	11.9 ₂₂	13.7 ₂₃₃	16.9 ₄₄₈	21.7 ₆₅₇	25.6 ₈₆₆	22.9 ₁₀₇₉	19.1 ₁₂₈₉	17.7 ₁₅₀₀
Specific heat (J/(kg•°C))	412 ₂₀	453 ₂₀₀	481 ₄₀₀	527 ₆₀₀	606 ₈₀₀	628 ₈₅₀	647 ₁₀₀₀	652 ₁₅₀₀
Laser power (W)	220	230	240	250	260			
Scanning speed (mm/s)	3	4	5	6				

Table 5 The chemical composition of Inconel 718 (wt.%)

Element	C	Cr	Ni	Mo	Ti	Nb	Mn	Si	Co	Fe
Content	0.05	18	52.5	2.96	0.9	5.15	0.25	0.22	0.66	Bal

from each deposition layer are randomly chosen as the training data, and 2000 samples from the remainder of the data set are chosen as the test data. We implement the PINN method on a 64-bit PC with a 6 GB GPU, a 6-core 2.9 GHz CPU, and 16 GB RAM. To make full use of the GPU parallel computing capability, the algorithm is coded based on the Tensorflow-GPU package.

4.1 Model performance evaluation

The error evaluation indices listed in Table 6 are used to evaluate the performance of the proposed temperature field prediction model using the PINN method. The root means squared error (RMSE), mean absolute error (MAE), and mean relative error (MRE) represent the prediction error. The closer their values are to 0, the smaller the error of the model. R^2 is used to evaluate the fitting ability of the model. The closer the value is to 1, the better the fitting ability of the model.

4.2 Temperature field prediction in single-layer DED using PINN method

Table 7 describes the performance of the model at a different time when the laser power is 240 W, and the scanning speed is 5 mm/s in single-layer DED. It turns out that the model has obtained a low prediction error, and the error value fluctuates little over time. Specifically, Fig. 5 shows the comparison between the predicted temperature field using PINN and the results of FEM on the upper surface of the deposition layer at different a time (2 s, 4 s, and 6 s). The temperature field of the DED parts evolves rapidly due to rapid heating and cooling. The predicted temperature contours of the DED

fabricated parts at different moments match those of FEM. For example, the RMSE of the predicted temperatures at different time (2 s, 4 s, and 6 s) is within 4.3 °C. The MRE at different moments (2 s, 4 s, and 6 s) is within 1.37%. Besides, the MAE of the predicted peak temperatures of the molten pool is within 3.8%. Therefore, the PINN-based temperature field prediction model possesses both high accuracy and good robustness over time in single-layer DED.

To depict more specific validation results of PINN, the predicted temperature distribution under different laser powers is presented in Fig. 6. With a higher laser power, the energy density injected into the molten pool is more intense, which results in higher temperatures of the molten pool. It can be seen that the predicted temperatures under different laser powers are also consistent with the result of FEM both in the tendency and the value. The maximum relative errors of predicted temperatures under 230 and 260 W are 3.8 and 3.2%, respectively.

The predicted temperature distribution along the laser scanning track under different scanning speeds (4 mm/s, 5 mm/s, and 6 mm/s) is illustrated in Fig. 7a. The peak temperature of the molten pool decreases with the increasing laser scanning speed that leads to the decreased energy absorbed by the molten pool. It turns out that the predicted temperatures are in good agreement with the result of FEM along the laser scanning track under different laser scanning speeds. The MAEs of the predicted temperatures are 2.63 °C, 3.13 °C, and 3.44 °C, respectively. The maximum relative errors of the predicted temperatures are 3.31%, 3.91%, 5.97%, respectively. Figure 7b depicts the predicted temperature gradients along the laser scanning track under the laser scanning speed of 5 mm/s. The absolute temperature gradient in the front of the molten pool is larger than that

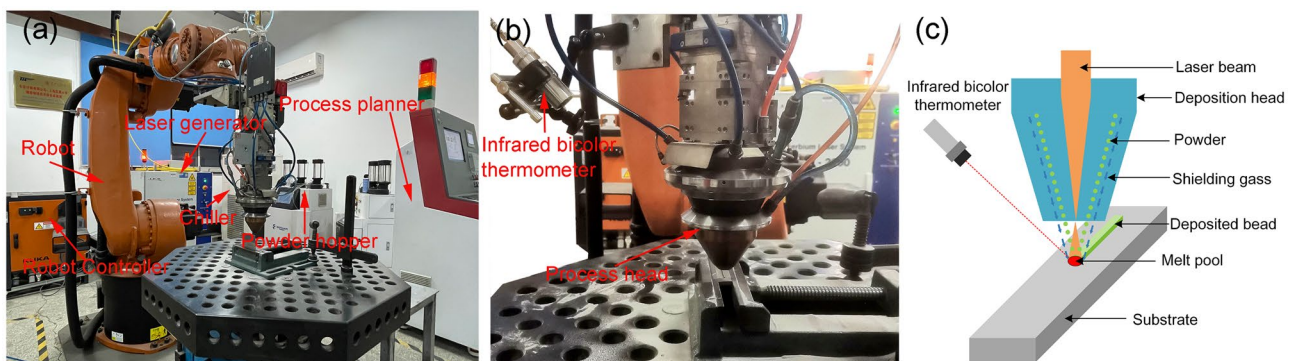
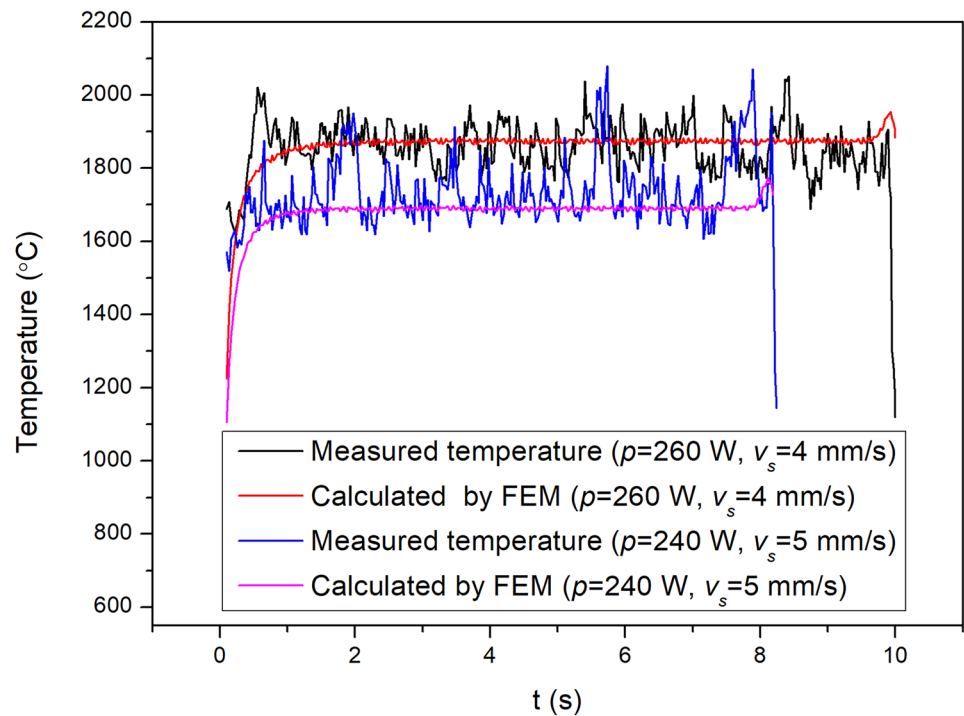


Fig. 3 The experiment system: **a** An overview of DED set up, **b** local view of the temperature measurement, and **c** schematic diagram of the temperature measurement

Fig. 4 Comparison of the measured and simulated temperatures of the molten pools



in the back due to the rapid heating by the moving laser beam. The predicted temperature gradients also agree with the result of FEM.

4.3 Temperature field prediction in multi-layer DED using PINN method

Temperature field prediction models for multi-layer DED have been established using PINN to validate the applicability of this physics-data hybrid method in more complicated DED cases. In multi-layer deposition, the laser scanning strategy and the substrate preheating are two important means to control the quality of the fabricated parts. The temperature fields under different deposition manners can result in different thermal stress distributions within fabricated parts, which are closely associated with cracking.

Here, three DED cases differing in the scanning strategy and preheating condition (Table 8) are investigated using the PINN method. The whole DED process lasts for 48 s (6 layers for bidirectional scanning and 3 layers for unidirectional scanning) under the laser power of 240 W and the scanning speed of 5 mm/s.

Figure 8 shows the predicted temperature fields on the cross-section (as defined in Fig. 9) during the DED process in the above three cases. It can be seen that increasing the layer number aggravates the heat accumulation and thus results in higher temperatures and larger heat-affected zones. The temperatures under the preheating condition are higher than those without preheating. Due to adequate cooling in the return phase of the laser process head, the temperatures under the unidirectional scanning are lower than those under bidirectional scanning. Especially, the visual comparison indicates that the predicted temperature fields using the PINN method are in good agreement with the results of FEM. In detail, as shown in Table 9, the MRE

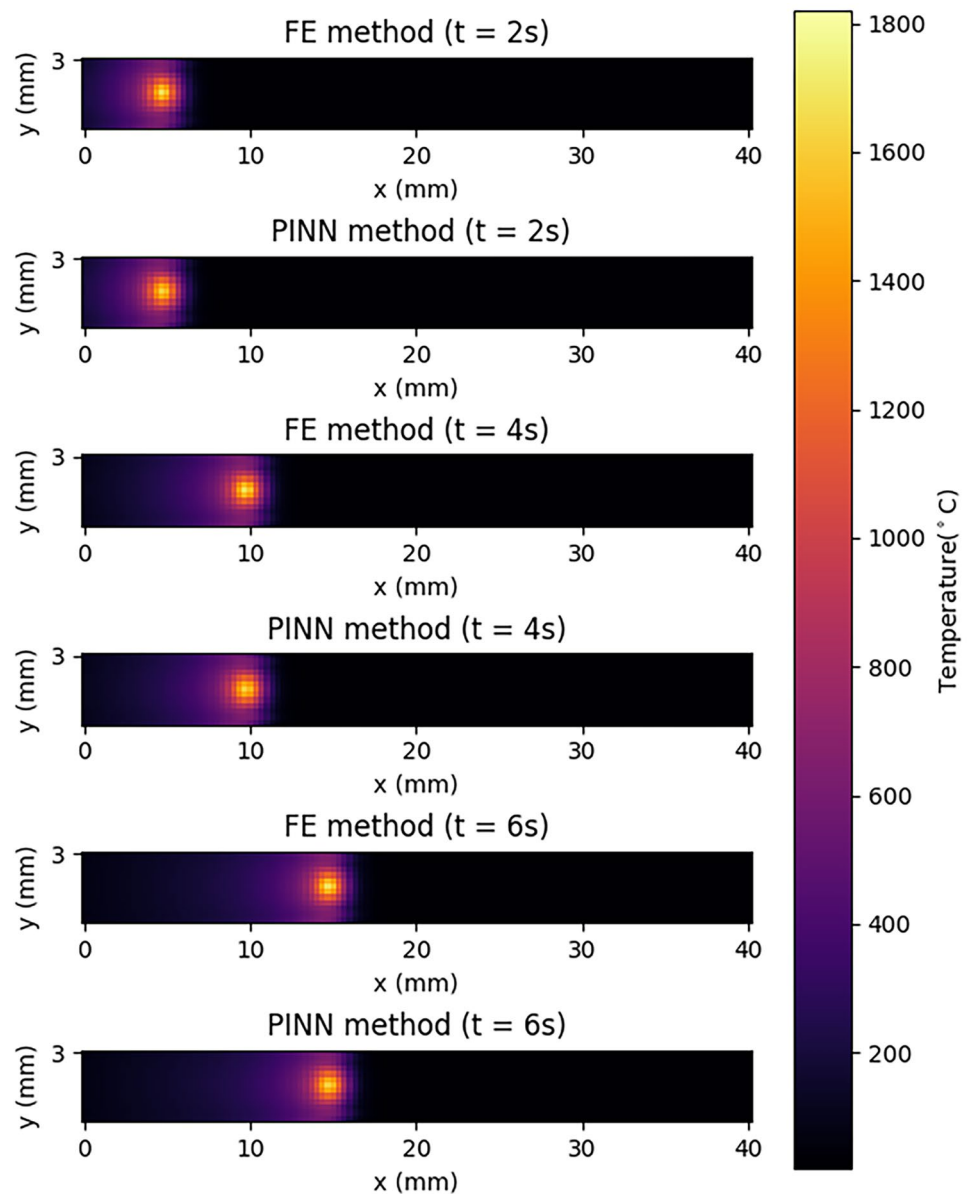
Table 6 Evaluation indices of the model

Indices	Description
Root mean squared error	$RMSE = \sqrt{\frac{1}{m} \sum_{i=1}^m (\hat{y}^{(i)} - y^{(i)})^2}$
Mean absolute error	$MAE = \frac{1}{m} \sum_{i=1}^m \hat{y}^{(i)} - y^{(i)} $
Mean relative error	$MRE = \frac{1}{m} \sum_{i=1}^m \left \frac{\hat{y}^{(i)} - y^{(i)}}{y^{(i)}} \right $
R^2	$R^2 = 1 - \frac{\sum_{i=1}^m (\hat{y}^{(i)} - \bar{y})^2}{\sum_{i=1}^m (y^{(i)} - \bar{y})^2}$

Table 7 Prediction performance at different time on single-layer DED

Case	RMSE (°C)	MAE (°C)	MRE	R^2
$t=2$ s	4.31	1.55	1.37%	0.9981
$t=3$ s	4.27	1.56	1.26%	0.9981
$t=4$ s	4.32	1.63	1.21%	0.9981
$t=5$ s	4.26	1.53	1.12%	0.9980
$t=6$ s	4.27	1.64	1.18%	0.9979
$t=7$ s	4.41	1.93	1.34%	0.9977
Total	4.32	1.62	1.28%	0.9980

Fig. 5 Comparison of the predicted temperature fields using PINN methods and FEM on the upper surface at $t=2$ s, 4 s, 6 s, respectively ($p=240$ W, $v_s=5$ mm/s)



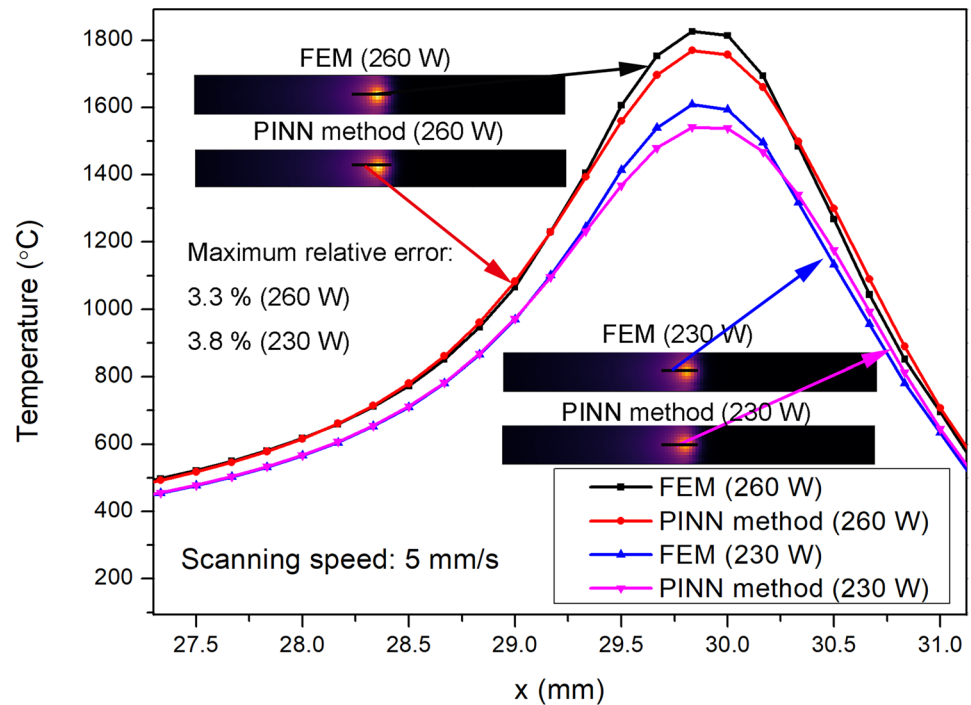
of the predicted temperatures for the three cases are 0.8%, 1.85%, and 3.5%, corresponding to R^2 of 0.99, 0.97, and 0.99, respectively. The results demonstrate the high accuracy and reliability of the PINN method in the temperature fields prediction of multi-layer DED.

Figure 10 depicts the predicted thermal history of the representative position (the point A in Fig. 9) on the deposition/substrate interface with and without preheating. The MAE of the predicted temperatures are 13.15 °C and 8.59 °C under no preheating and preheating conditions, respectively. Since the substrate has been preheated, the peak temperature is higher than that without preheating. This is in good agreement with the result of the previous work [41]. Besides, the difference between the maximum and minimum temperatures in each thermal cycle under the preheating condition is smaller than

that under no preheating condition, thus the deposition/substrate interface is thought to experience minor thermal stress and distortion by preheating the substrate [42, 43].

Figure 11 illustrates the predicted thermal history of the representative position (the point A in Fig. 9) on the deposition/substrate interface under unidirectional scanning and bidirectional scanning. It can be seen that the predicted thermal history using the PINN method is consistent with that using FEM in both tendency and value. The MAEs of the predicted temperatures by unidirectional scanning and bidirectional scanning are 13.15 and 7.38 °C, respectively. In contrast to the continuous heating under bidirectional scanning, the heating in the return phase of unidirectional scanning is suspended without imposed laser energy. Each deposited layer will be cooled

Fig. 6 The predicted temperatures under different laser powers



adequately in the return phase. Thus, during each thermal cycle, as shown in Fig. 11, the deposition/substrate interface undergoes relatively large temperature oscillations under unidirectional scanning compared to those under bidirectional scanning and higher residual stresses will be produced [44].

4.4 Experimental validation

In order to validate the temperatures predicted by the PINN-based model, three different DED experiments of Inconel 718 were conducted. The multi-layer DED is widely used

in the manufacturing of Inconel 718, where different laser scanning strategies are commonly adopted to control the temperature distribution. Specifically, the bidirectional scanning and unidirectional scanning (see Fig. 9) were experimentally conducted to verify the above predictions from the PINN method. In addition, another DED experiment was performed under the bidirectional scanning with substrate preheating (heating the substrate to 400 °C by the laser prior to deposition). All the experiments above were conducted under the laser power of 240 W, the laser scanning speed of 5 mm/s, and the powder feeding speed of 7.6 g/min, which are appropriate parameters to yield good forming quality of

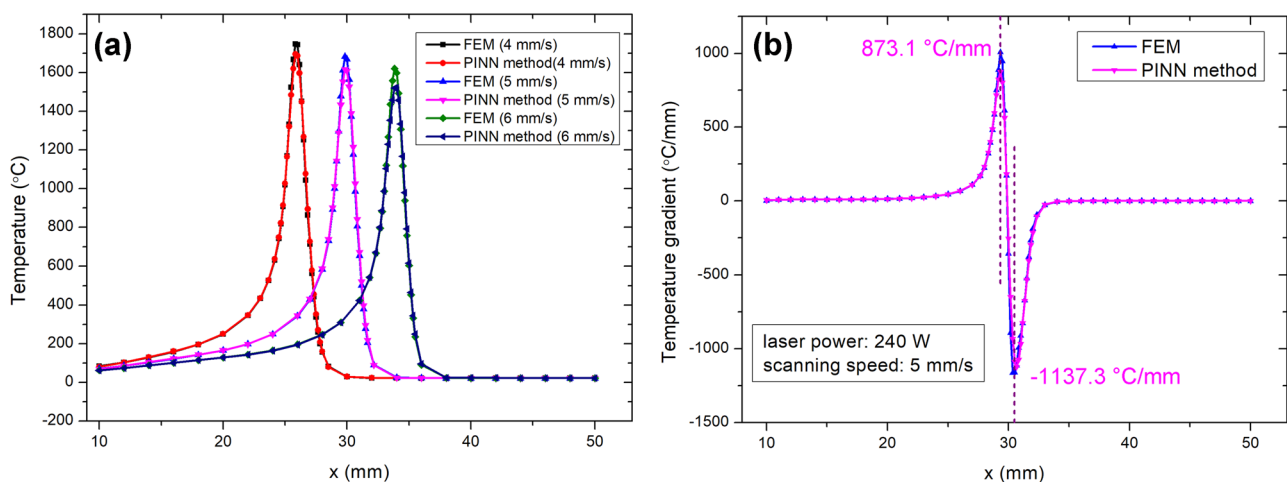


Fig. 7 Temperature distribution along the scanning track at $t = 4$ s: **a** predicted temperatures, and **b** predicted temperature gradients

Table 8 Processing conditions of the three cases (240 W, 5 mm/s)

Cases	Scanning strategy	Preheating condition	Number of deposited layers
1	Bidirectional	No Preheating	6
2	Bidirectional	Preheating	6
3	Unidirectional	No preheating	3

Inconel 718. Meanwhile, the infrared bicolour thermometer (see Fig. 3) was used to collect the molten pool temperature in real-time.

Figure 12a compares the predicted temperatures and the experimentally measured temperatures under the bidirectional scanning. It can be seen that the predicted temperatures are consistent with the measured temperatures of the molten pool over time. The MRE is 4.83% in consideration of the fluctuations in measurements. Figure 12b compares the predicted temperatures and the experimentally measured temperatures under the condition that the substrate has been preheated to 400 °C prior to DED. The temperatures of the molten pool rise with the increase of deposited layers under the preheating condition, which demonstrates the

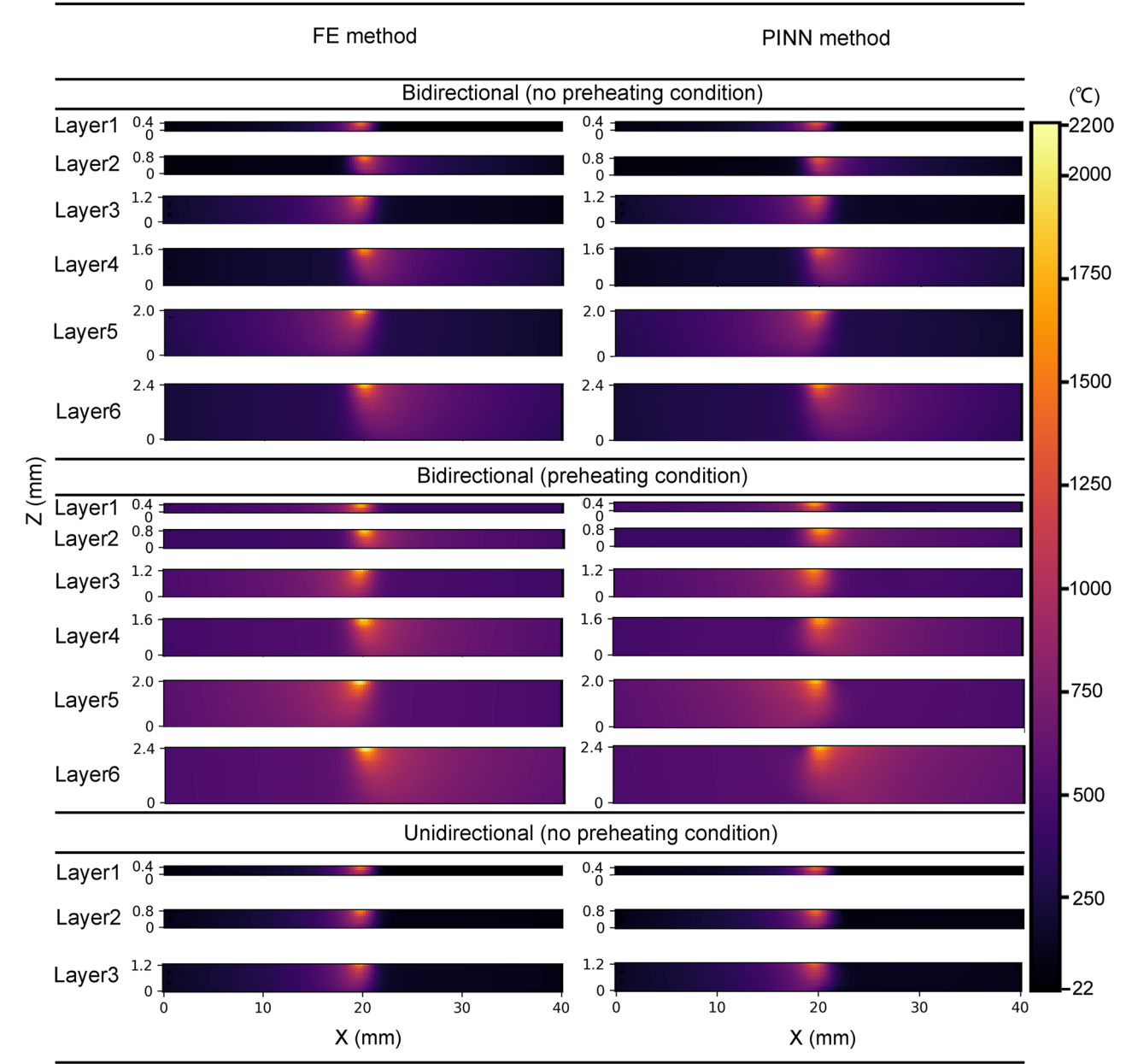


Fig. 8 Comparison between the results of FEM and the results of PINN method on the cross section

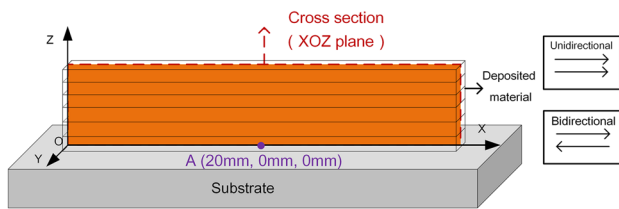


Fig. 9 Schematic diagram of multi-layer DED

heat accumulation effect. The predicted temperatures agree with the actual temperatures in both tendency and value. The MRE of the predicted temperatures is 2.85%. Figure 12c compares the predicted temperatures and the experimentally measured temperatures under the unidirectional scanning. In this case, the heat accumulation effect is not obvious due to the sufficient cooling during the return phase without laser heating. The MRE of the predicted temperature is 3.21%. The experimental validation results above indicate that modeling the temperature field during DED using PINN is effective and reliable.

5 Discussion

5.1 The physical part in the loss function of PINN

In examining the residual of the heat conduction PDE in the loss function of PINN, we have also trained and tested a PINN-based temperature field prediction model for 316L stainless steel. This validates the applicability of the proposed physics-data hybrid method to metals other than Inconel 718 and hence demonstrates the generalizability of the proposed method. Given that the trained model is constrained by the laws of heat transfer, the residuals of the heat conduction PDE (Eq. 10) in the loss function of the trained models are expected to approach 0. To verify this, 1200 sample points (p , v_s , t , x , y , and z) are randomly chosen from each of the test sets of the two metals as the input of the trained models. The predicted temperatures (T) are computed through forward propagation, and then the partial derivatives of T with respect to x , y , z , and t are computed by the AD function that is imported from the Tensorflow library. Finally, the residual of the PDE (Eq. 10) in the loss function can be obtained. Figure 13 shows the computed residual values of the PDE (R_T) with randomly chosen samples. It turns out that the residual values for both Inconel 718

and 316L stainless steel are close to 0. The mean absolute values of R_T for c are 0.0016 and 0.0017, respectively. Good convergence of the PDE residual indicates that the PINN-based model is constrained by the laws of physics after iterative backpropagations.

The actual diffusion coefficient of a material is often difficult to be determined by an experimental method. Besides, it is inadvisable to replace α (as defined in Eq. 10) with the measured diffusion coefficient directly because the physical dimensions of the input data have been eliminated by the normalization operation that scales the input data to a specific range. In this paper, we treat α as an undetermined variable that can be inferred inversely based on the available data of the temperature distribution in the learning process. As shown in Fig. 14, the value of α is constantly updated together with the training of the PINN-based model through iterative backpropagations until the default convergence criteria of the L-BFGS optimizer is reached. Figure 15 shows the relationship between the output of $\partial_t T$ and $\nabla^2 T$ (see Eq. 10) after implementing the forward propagation on the trained models with randomly chosen sample points from the test set. It turns out that the scatters conform to the linear distribution and the slopes of the fitted lines represent the values of α for Inconel 718 and 316L stainless steel after implementing the forward propagation with the test data set, respectively. The values of α on the test data set are almost consistent with the learned α shown in Fig. 14. The linear relationship between $\partial_t T$ and $\nabla^2 T$ on the test data further demonstrates that the trained model is constrained by the laws of physics in the current study. The performance of PINN mentioned in the previous section indicates that the strategy of dynamically updating α is effective and can contribute to high prediction accuracy. Thus, the main structure of physical terms in the loss function of the proposed model is informed by the heat conduction PDE, but the involved parameter α is treated as a variable that will be learned through iterative backpropagations in the learning process. The PINN-based model hybridizes the physical law and data via the following two aspects. Firstly, the loss function consists of the mean squared error (MSE) of the discrepancy of the predicted temperatures and the labeled temperatures and the MSE of the PDE residual value. Secondly, the definition of the physical terms in the loss function is informed by the laws of heat transfer, while the corresponding parameter (α) in the physical terms is determined by the data-driven method.

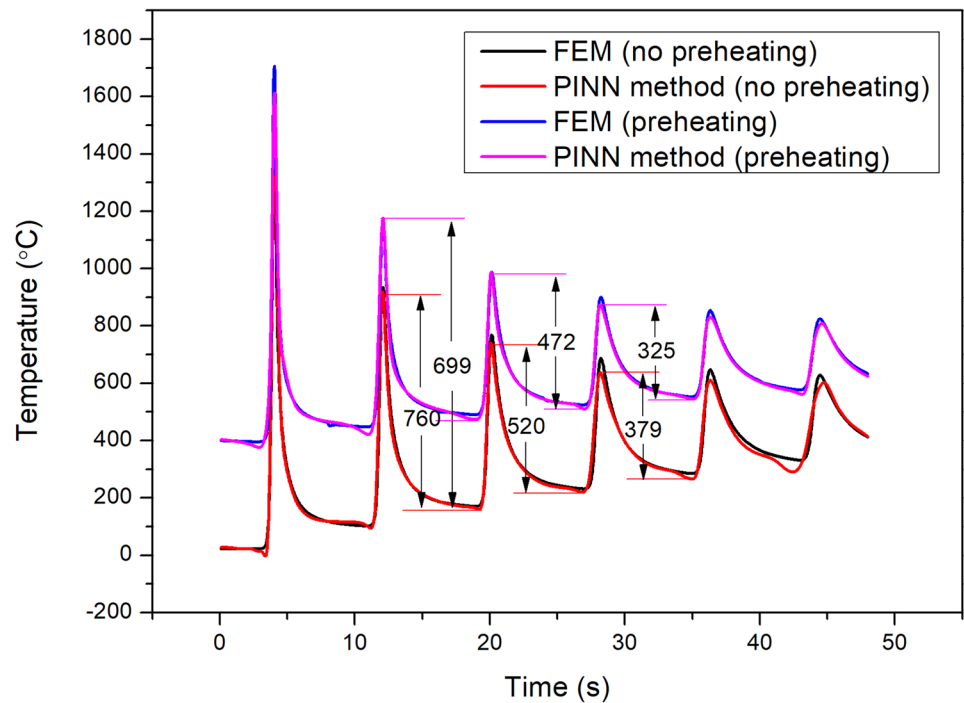
5.2 Comparison with data-driven methods

In order to further investigate the performance of the PINN-based model, comparisons are made with other models based on data-driven methods, i.e., DNN, LSTM, and XGBoost. DNN is a typical data-driven machine learning algorithm

Table 9 Prediction performance in different cases of multi-layer DED

Cases	RMSE (°C)	MAE (°C)	MRE	R ²
1	10.87	4.42	0.8%	0.99
2	28.30	12.85	1.85%	0.97
3	18.05	9.46	3.5%	0.99

Fig. 10 Comparison between the thermal history of point A obtained by FEM and by the PINN method under preheating condition and no preheating condition

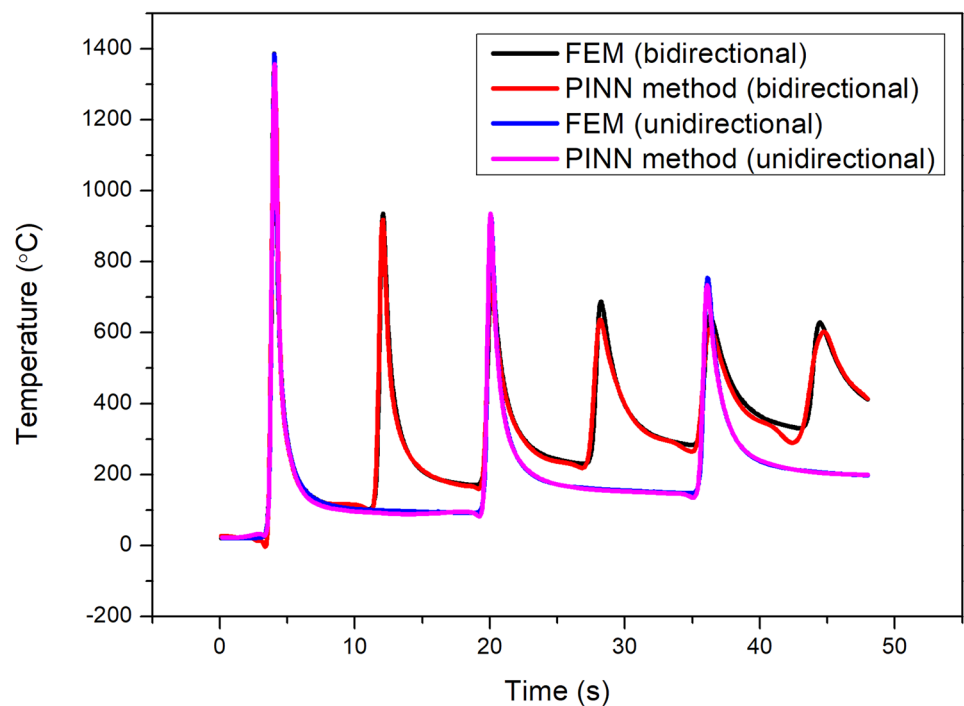


which updates parameters through forward propagation and backpropagation. Then the gap between the predicted value and the labeled data will be narrowed after repeated iterations. LSTM with gate regulation (forget gate f_t , update gate u_t , output gate o_t , see Fig. 16) is a solution to the gradient vanishment that occurs in RNN by selectively transferring valuable information in forward propagation. XGBoost is an

improved gradient boosting decision tree (GBDT) algorithm and has gained great popularity due to its high computational efficiency and excellent performance [45]. In XGBoost, each tree works as a weak base learner to fit the residual of the previous learner so as to boost the model performance.

We have investigated the impact of the size of the training data on the prediction accuracy for the above models. The

Fig. 11 Comparison between the thermal history of point A obtained by FEM and by the PINN method under different scanning strategies



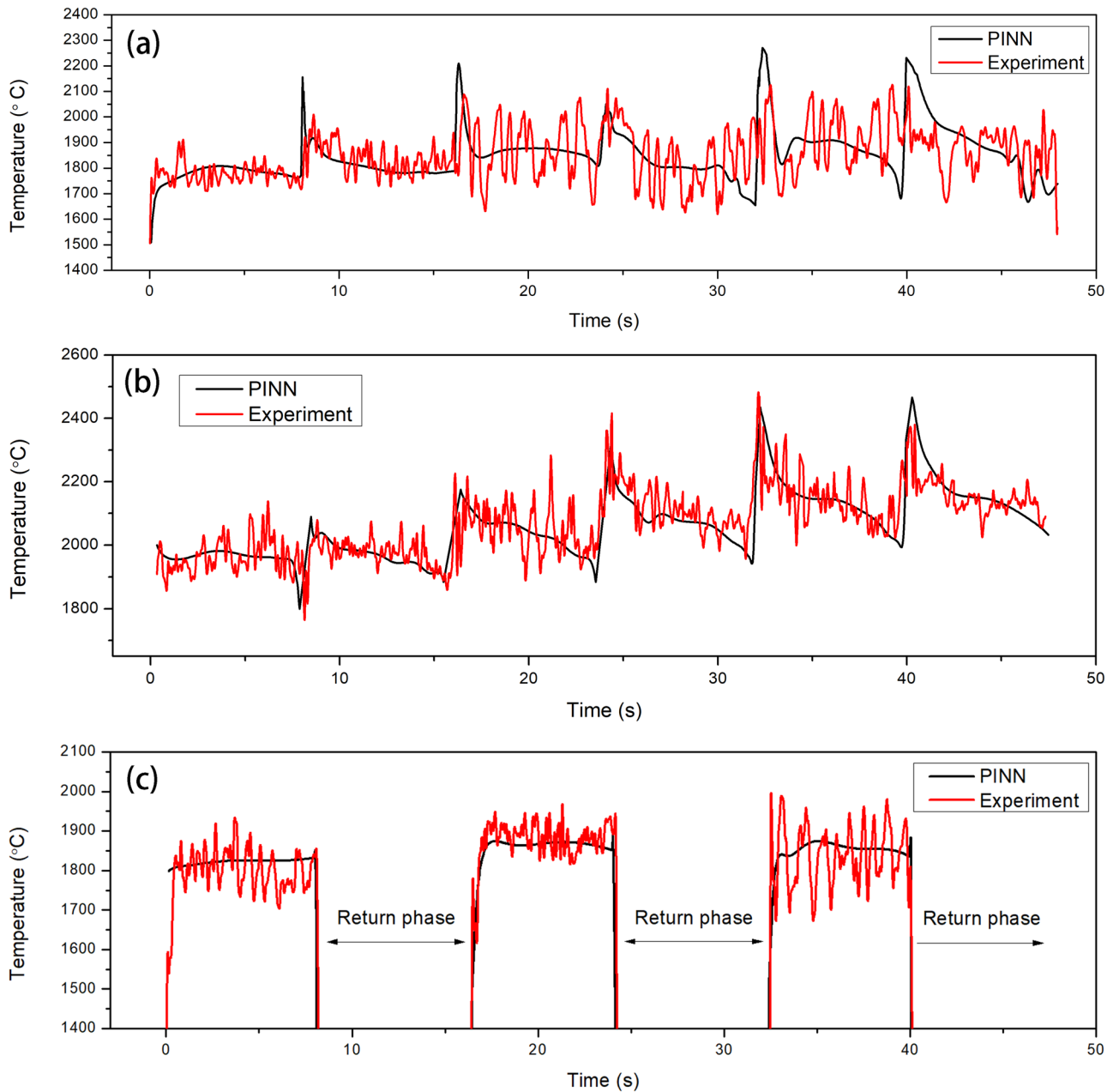


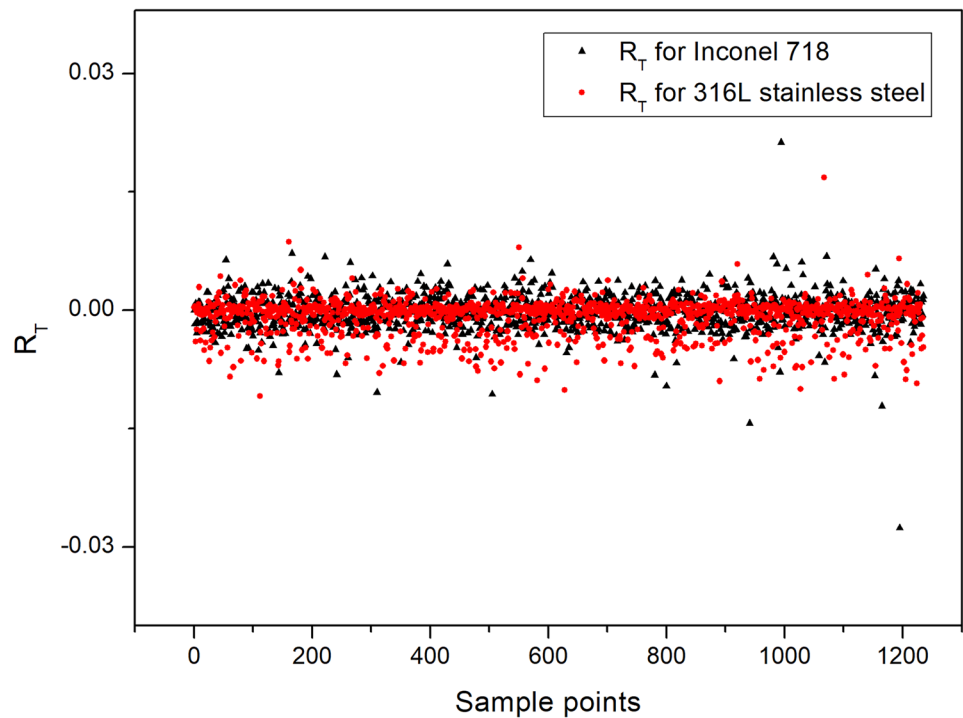
Fig. 12 Predicted and measured molten pool temperature: **a** bidirectional scanning, **b** under preheating condition, and **c** unidirectional scanning

PINN, DNN, LSTM, and XGBoost are trained with multiple sets of input data and then are validated with the same test data. The formula based on the relative L_2 error is used to evaluate the prediction accuracy [46] on the test data set:

$$\text{Accuracy} = 1 - \frac{\sqrt{\sum_{i=1}^m (\hat{y}^{(i)} - y^{(i)})^2}}{\sqrt{\sum_{i=1}^m (y^{(i)})^2}} \quad (18)$$

where $\hat{y}$ is the predicted value and y is the labeled value. The prediction accuracy of the data-driven model largely depends on the number and representative quality of the training data. To train a reliable model, the labeled data that covers most engineering situations are required [24]. Figure 17 depicts the influence of the training data number on the prediction accuracy of the four models. A common feature for the PINN, LSTM, DNN, and XGBoost based models is that a larger training set can result in higher prediction accuracy. Particularly, the XGBoost based model shows a more obvious

Fig. 13 The residuals of PDE estimated with randomly chosen input data for the trained models of Inconel 718 and 316L stainless steel



dependence on the amount of training data in the temperature prediction of DED. When the number of training data is less than 2000, the prediction accuracy of XGBoost is the lowest among the four algorithms. When the number of training data reaches 80,000, the prediction accuracy of the XGBoost can reach 96%, which is higher than that of LSTM but still lower than that of PINN. The prediction accuracy of

the PINN-based model is higher and has less fluctuation than those of LSTM, XGBoost, and DNN on the whole, and this is remarkable as the number of training data is less than 10,000. Therefore, the PINN-based model is capable of achieving a higher accuracy when massive data are not available.

Further, we assess the minimal number of training data that are required for the four modeling methods to

Fig. 14 The updating process of α during learning

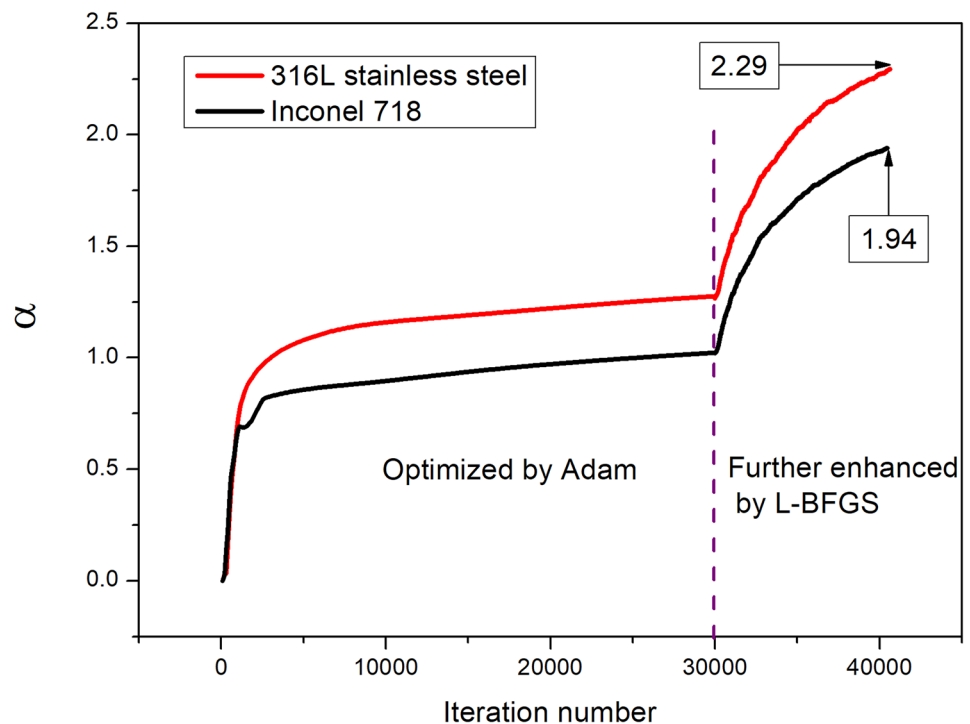
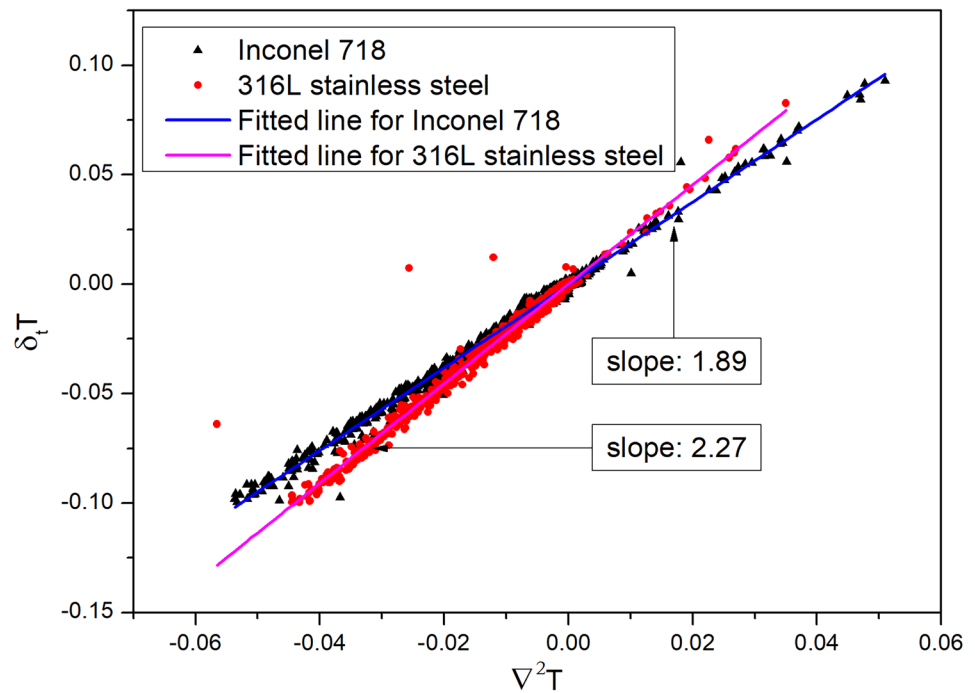


Fig. 15 Predicted $\partial_t T$ versus $\nabla^2 T$ after implementing the forward propagation on the trained model with the test data



achieve identical accuracy. As shown in Fig. 18, the number of training data required for PINN is 12% of that for DNN to achieve the prediction accuracy of 80%. Specifically, 5000 labeled data are required to train DNN, 2000 labeled data are required to train LSTM, and only 600 labeled data are required to train PINN. To achieve the prediction accuracy of 90%, 20,000 labeled data are required to train DNN and XGBoost, 8000 labeled data are required to train LSTM, and 4000 labeled data are required to train the PINN-based model. This means that PINN can achieve identical prediction accuracy with 20% of training data required for DNN and XGBoost. Therefore, to achieve the desired prediction accuracy, the labeled data required by PINN is much less than that of LSTM, XGBoost, and DNN. Because the loss function of

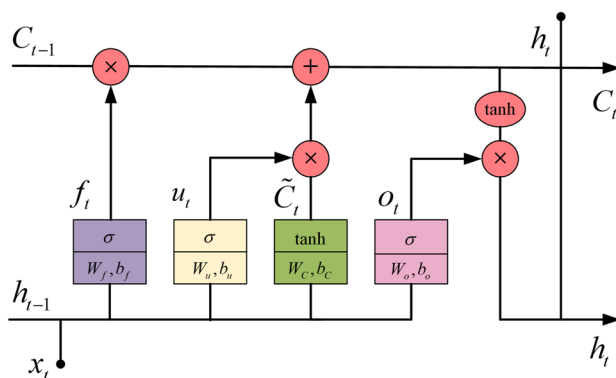


Fig. 16 Schematic diagram of LSTM

PINN consists of physical terms and data terms (Eq. 11), the updated model parameters in each iteration make the predicted values not only approaching the real value but also approaching the feasible solution of the partial differential equations. Therefore, the physics-data hybrid model based on PINN is useful when the cost of collecting training data by experiments or numerical simulations is expensive but the corresponding physical process can be defined by some physical PDEs.

In order to evaluate the performance of the proposed PINN model more comprehensively, we compared the computational efficiencies of the above four models for achieving an identical prediction accuracy of 90%. Table 10 shows the comparison results. As a tree boosting algorithm, XGBoost has the highest computational efficiency in model training and prediction, but it requires more training data than PINN to achieve the target accuracy. Compared with DNN, PINN requires a great number of additional automatic differentiation operations to calculate the partial derivative of the loss function, so the amount of calculation, hence the training time, increases greatly. However, the difference in the mean prediction time costs for PINN and DNN is marginal. It is apparent that LSTM is inferior to the other three models with respect to the highest mean prediction time cost. Overall, the PINN-based model is capable of achieving higher prediction accuracy with less data and computation time of about several minutes, which compares favorably with XGBoost, DNN, and LSTM in consideration of the high training-data collection cost in DED manufacturing.

Fig. 17 The accuracy of the temperature field prediction models with different number of training data

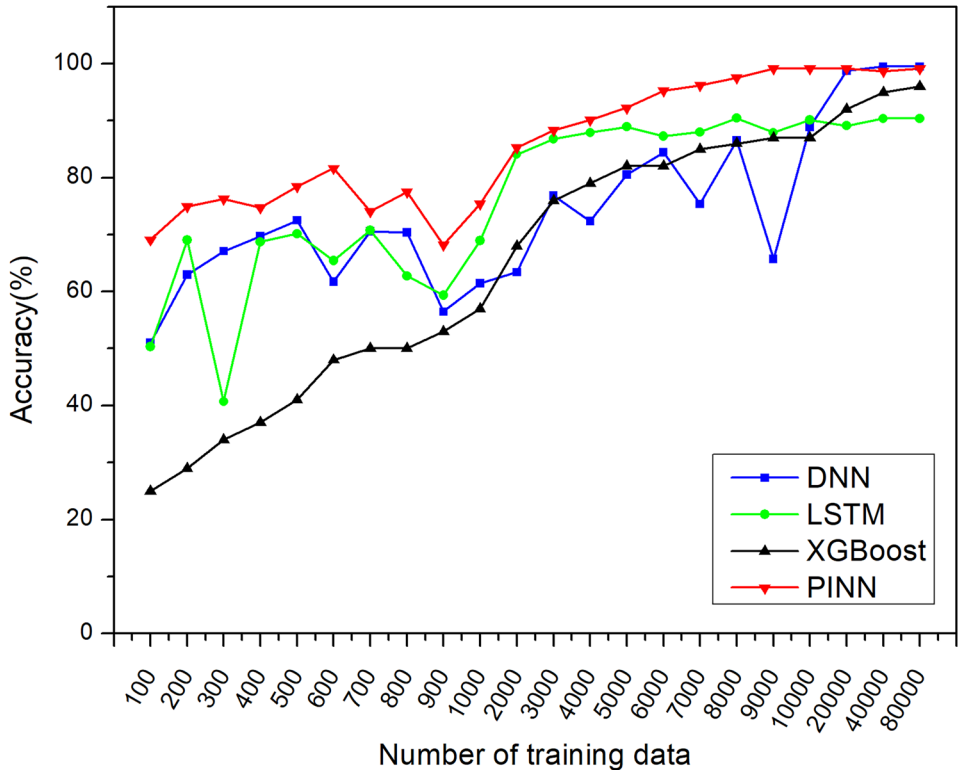


Fig. 18 The required minimal number of training data for achieving the given prediction accuracy

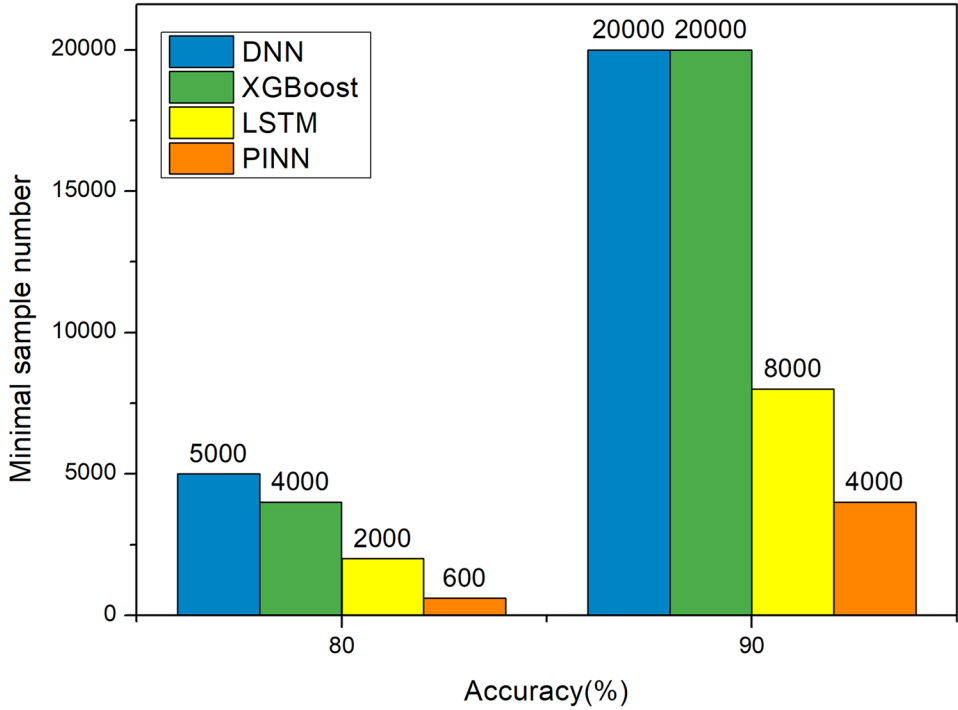


Table 10 Computational efficiencies of the four models for achieving the accuracy of 90%

Methods	DNN	LSTM	XGBoost	PINN
Number of training data needed	20,000	8000	20,000	4000
Training time cost (s)	182.06	235.19	10.18	622.41
Mean prediction time cost (ms)	0.18	1.03	0.01	0.20

6 Conclusion

PINN has been employed to predict the 3D temperature field in both single-layer and multi-layer DED processes. The heat conduction partial differential equations are embedded into the loss function of the neural network, which allows the model to be trained by data and to be constrained by the laws of physics simultaneously. After discussion and analysis, the following conclusions are drawn:

1. The physics-data hybrid machine learning method is capable of predicting the 3D temperature field accurately in DED under different processing conditions.
2. The PINN-based model is more explainable with the physical laws than traditional machine learning models. The main structure of the physical term in the loss function is informed by the laws of heat transfer, and the immeasurable parameter in the physical term is updated iteratively by the data-driven method. The extra embedded PDE guides the predicted value to approach the feasible solution in the learning process.
3. Compared with data-driven machine learning methods including DNN, LSTM, and XGBoost, PINN can achieve high prediction accuracy with significantly less data due to the additional constraint enforced by the physical laws.

This paper focuses on demonstrating the ability of PINN to model the temperature field, and the training data are generated by numerical simulation. Future work would involve measuring the temperature of every point both on the surface and inside the fabricated parts and training the model with the measured data to further improve the temperature predicting accuracy in DED. Also, future work would make the physics-data hybrid approach applicable for online weld pool monitoring.

Author contribution Jibing Xie: investigation, methodology, coding, analyzing, writing—original draft. Ze Chai: writing—review and editing. Xukai Ren: writing—review and editing. Luming Xu: experimentation. Sheng Liu: data analysis. Xiaoqi Chen: conceptualization, funding acquisition, writing—review, and revision.

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Availability of data and material The data that support the findings of this study are available from the corresponding author upon reasonable request.

Code availability All the codes of this study are available from the corresponding author upon reasonable request.

Declarations

Ethical approval Not applicable.

Consent to participate The manuscript has been read and approved by all authors.

Consent for publication Not applicable.

Competing interests The authors declare no competing interests.

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